

The Polyhedron Zoo

ENRICHMENT PROJECT

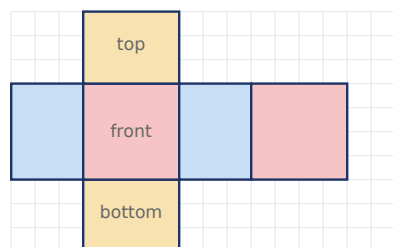
Day 17 Enrichment · MathCraft CPA · Grade 6 · Professor Xavier Honablue, M.Ed.

Materials: graph paper ($\frac{1}{4}$ -inch squares) · ruler · scissors · tape · colored pencils. **Standards used:** 6.G.A.4 (nets and surface area), 6.G.A.1 (area of triangles), 6.NS.B.3 (decimals). **Advanced Outlook:** Euler's Formula and the surface-area-to-volume ratio go beyond 6th grade and are not tested. **Rule:** ruler for every line; label every face with its dimensions before you cut.

Build 1 · A rectangular prism from your own net

Draw the net below on graph paper at 1 square = 1 inch, so the finished box is 4 in $\times$ 3 in $\times$ 4 in. Color the three matching pairs of faces in three different colors — matching faces get the same color. Cut, fold, and tape.

1 square = 1 inch · this net folds into a 4 $\times$ 3 $\times$ 4 box



1. Before folding, label every face with its dimensions and its area.
2. Find the surface area two ways: add all six face areas, and use $SA = 2(lw) + 2(lh) + 2(wh)$. They must agree.
3. Count on your finished model: V (corners) = ___, E (edges) = ___, F (faces) = ___. Compute $V - E + F$.

Build 2 · A triangular prism

Design a net for a triangular prism 6 in long whose ends are right triangles with legs 3 in and 4 in (hypotenuse 5 in). That is two triangles plus three rectangles — one rectangle per side of the triangle.

4. Sketch the net on graph paper, label every piece, then build it.
5. Find the surface area: two triangles + three rectangles.
6. Count V , E , F on your model and compute $V - E + F$.

Build 3 · A square pyramid

Design a net for a pyramid with a 4 in $\times$ 4 in square base and four identical triangular sides with a slant height of 5 in (the height of each triangle measured along the face).

7. Build it, then find the surface area: the square base + four triangles.
8. Count V , E , F and compute $V - E + F$.

Zoo Report

9. Fill in the table for all three solids, then write one sentence about what you notice in the last column.

Solid	Surface area	V	E	F	$V - E + F$
-------	--------------	-----	-----	-----	-------------

Rectangular prism ($4 \times 3 \times 4$)					
Triangular prism (3-4-5, length 6)					
Square pyramid (base 4, slant 5)					

Scaling Experiment (Advanced Outlook)

10. Build a second cube-shaped box with double the edge length of a 2 in cube (so 4 in). Find each one's surface area and volume, then the ratio $SA \div V$. Which box has more skin for its size? Write one sentence explaining what that means for an ice cube melting in a glass of water.

Display & Presentation

11. Set your three solids on a card labeled "The Polyhedron Zoo," with an index card for each one showing its net sketch, its surface area with work, and its $V - E + F$ score.

Scoring Rubric (20 points)

Category	4 pts	2 pts	0 pts
Nets drawn accurately with a ruler and to scale	All three	One or two	None
Faces labeled with dimensions before cutting	All solids	Some	None
Surface area computed correctly, work shown	All three	Some	Missing
Models built, folded, and taped cleanly	All three	Some	None
Zoo report table + noticing sentence	Complete and correct	Partly complete	Missing

Teacher Answer Key

Build 1 ($4 \times 3 \times 4$ box): $SA = 2(4 \times 3) + 2(4 \times 4) + 2(3 \times 4) = 24 + 32 + 24 = 80$ sq in. $V = 8$, $E = 12$, $F = 6 \rightarrow 8 - 12 + 6 = 2$.

Build 2 (triangular prism, legs 3 and 4, length 6): two triangles $2 \times (\frac{1}{2} \times 3 \times 4) = 12$; rectangles $(3 + 4 + 5) \times 6 = 72$; $SA = 84$ sq in. $V = 6$, $E = 9$, $F = 5 \rightarrow 6 - 9 + 5 = 2$.

Build 3 (square pyramid, base 4, slant height 5): base $4 \times 4 = 16$; four triangles $4 \times (\frac{1}{2} \times 4 \times 5) = 40$; $SA = 56$ sq in. $V = 5$, $E = 8$, $F = 5 \rightarrow 5 - 8 + 5 = 2$.

Scaling (item 10): 2 in cube — SA 24 sq in, V 8 cu in, ratio 3.0. 4 in cube — SA 96 sq in, V 64 cu in, ratio 1.5. The smaller cube has more surface for its size, which is why crushed ice melts faster than one big block.

Noticing sentence (item 9): every solid gives $V - E + F = 2$ — Euler's Formula.

Where this leads: [NRICH \(University of Cambridge\) — Euler's Formula](#) · [Wolfram MathWorld — Pick's Theorem](#) · [GeoGebra — fold and unfold a prism](#) · [CMU CS Academy — code your nets](#).